

C^∞ smoothness of the fluctuation function and its ladder extension

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Abstract

The foundation unit for the extended ladder: the fluctuation function S_λ is C^∞ in a , the lowering operator b preserves C^∞ , and hence the recursive extension $S_{\mu-k/2} = \text{Sneg } \beta \mu k$ is C^∞ for base order $\mu > 0$. This supplies the differentiability inputs (`Differentiable` of `Sneg` and of its derivative) that the Weyl commutator `ladder_commutator` demands, unblocking the extended raising relation. Zero `sorry/axiom`.

1 Setting

The ladder algebra of grammar §4 needs C^2 regularity to apply $[b, b^\dagger] = 1$: the commutator proof differentiates f twice. The extension $S_{\mu-k/2}$ is built by applying the lowering operator $b = 2\beta^{-1/2}\partial_a - \beta^{1/2}a$ repeatedly, so establishing that each stage is smooth is a prerequisite for the *opposite* (raising) direction. The engine is the derivative ladder $S'_\lambda = \beta S_{\lambda+1/2}$ (unit 2): the family of fluctuation functions is closed under a -differentiation, so *every* iterated derivative exists and is continuous.

2 The things formalised

```
theorem contDiff_fluctuation ( lam : ) (h : 0 < ) (hlam : 0 < lam) :
  ContDiff ( : ω) (fun a => fluctuation lam a)
theorem contDiff_lowerOp ( : ) (f : → ) (hf : ContDiff ( : ω) f) :
  ContDiff ( : ω) (lowerOp f)
theorem contDiff_Sneg ( : ) (h : 0 < ) (h : 0 < ) (k : ) :
  ContDiff ( : ω) (fun a => Sneg k a)
theorem differentiable_Sneg ... -- Differentiable (Sneg k)
theorem differentiable_deriv_Sneg ... -- Differentiable (deriv (Sneg k))
```

3 Proof strategy

Smoothness of S_λ is by an auxiliary finite-order induction `contDiff_nat_fluctuation`: for every $n : \mathbb{N}$ and $\lambda > 0$, `ContDiff  $\mathbb{R}$   $n$   $S_\lambda$` . The base is `contDiff_zero` (continuity, from `hasDerivAt_fluctuation`); the step uses `contDiff_succ_iff_deriv` with $\partial_a S_\lambda = \beta S_{\lambda+1/2}$ staying in the family, so the inductive hypothesis at $\lambda + 1/2$ closes it. Then `contDiff_infty` lifts $\forall n$ to ∞ . Smoothness of bf is compositional: $\partial_a f$ is C^∞ (`contDiff_infty_iff_deriv`), and bf is a constant-scaled sum of $\partial_a f$ and $a f$. Smoothness of `Sneg` is then a one-line induction on k using `contDiff_lowerOp`.

4 Roadblocks and resolutions

The `ContDiff` index has migrated to `WithTop N $\infty$` , adding an analytic grade $\omega = \top$ above ∞ . Three frictions followed:

- `contDiff_succ_iff_deriv` gained a clause $\uparrow n = \omega \rightarrow \text{AnalyticOn}$. For finite n this is vacuous, discharged by `intro h`; `exact absurd h` (by `simp`).
- Coercions $\mathbb{N} \rightarrow \mathbb{N}_\infty \rightarrow \text{WithTop } \mathbb{N}_\infty$ fight the finite- n induction; `Nat.cast_succ` plus `exact_mod_cast` bridge them.
- `.differentiable le_top` *fails* — `le_top` yields $\leq \omega$, not $\leq \infty$, and `exact_mod_cast le_top` also fails. The working incantation for “ $1 \leq \infty$ ” is simply `(by norm_num)`.

Also: `ContDiff.const_smul` produces $c \bullet x$, not $c * x$; use `contDiff_const.mul` to land the multiplicative form the downstream `deriv` rewrites expect.

5 What was Mathlib, what was new

Mathlib supplied the `ContDiff` calculus (`contDiff_succ_iff_deriv`, `contDiff_infty`, `contDiff_infty_iff_deriv`, `contDiff_zero`, `contDiff_const`, `contDiff_id`). New is the observation that the fluctuation family is differentiation-closed and the resulting clean induction, plus the smoothness-preservation of the lowering operator.

6 Lessons

On the current pin, `ContDiff`’s top index is `WithTop N $\infty$` ; write finite regularity through an explicit $\mathbb{N}$ -induction and lift with `contDiff_infty`, and reach for `(by norm_num)` rather than `le_top` when you need $1 \leq \infty$ or $2 \leq \infty$.

7 Follow-ups

With `differentiable_Sneg` and `differentiable_deriv_Sneg` in hand, the extended raising relation and number operator become provable (the very next unit).